Customer Sentiment Analyzer

Project Report

Python | NLP | Pandas | Matplotlib

1. Project Overview

This focuses on using Natural Language Processing (NLP) and Python to create a Customer Sentiment Analyzer. Without having to manually read each review, the primary goal was to automatically categorize customer reviews into three sentiment categories: positive, neutral, and negative. This enables companies to make data-driven decisions and rapidly comprehend client happiness.

2. Tools & Technologies Used

Tool/Library	Purpose
Python	Core programming language
Pandas	Data loading, cleaning, and manipulation
TextBlob	NLP library for sentiment polarity analysis
Matplotlib	Data visualization and charts
WordCloud	Visual representation of frequent words
Google Colab	Coding environment

3. Dataset

A structured dataset of 300 customers reviews was used for this project, covering 5 kitchen appliance products:

- Kitchen Mixer
- Coffee Maker
- Toaster
- Blender
- Microwave

Each review contained the review and a star rating (1 to 5). The dataset was created with realistic positive, neutral and negative reviews to stimulate real world customer feedback.

4. Methodology

Step 1: Library Installation: Use pip to install the necessary Python libraries, including Pandas, Matplotlib, TextBlob, and Wordcloud.

Step 2: Dataset Creation: 300 reviews of five kitchen products with star ratings ranging from 1 to 5.

Step 3: Sentiment Labeling: Each review was given a rating of 4-5 stars for positive, 3 stars for neutral, and 1-2 stars for negative.

Step 4: NLP Analysis: To analyze review content and determine polarity scores for automatic sentiment prediction, use the TextBlob NLP package.

Step 5: Accuracy Calculation: Model accuracy (66.67%) was determined by comparing TextBlob projected sentiments with score-based labels.

Step 6: Visualizations: Using Matplotlib, I made pie charts, bar charts, a WordCloud, and a sentiment comparison by product.

Step 7: Business Summary: A final summary table with the total number of reviews, average score, and sentiment breakdown for each product was created.

5. Results & Findings

Product	Total Reviews	Avg Score	Positive	Negative
Blender	55	3.53	42	13
Coffee Maker	68	3.13	46	22
Kitchen Mixer	70	2.76	32	38
Microwave	60	2.88	35	25
Toaster	47	2.70	25	22

Key Findings :

- 200 out of 300 reviews were correctly classified, yielding a model accuracy of 66.67%.
- The best-performing product is the blender (average rating: 3.53, most positive evaluations).
- The toaster (average rating of 2.70, requires improvement) is the worst performing product.
- Out of 70 reviews, Kitchen Mixer had the most unfavorable ones (38).
- The overall sentiment split for all goods was 120 negative and 180 positive.

6. Conclusion

The project effectively illustrated how customer sentiment analysis can be automated using NLP and Python. As might be expected for a simple rule-based NLP model, TextBlob's polarity rating obtained 66.67% accuracy. The summary table and visualizations offer concise, useful information for making business decisions. In the future, accuracy can be increased by training on larger real-world datasets and utilizing sophisticated models like VADER or BERT.

7. Future Improvements

- For greater accuracy, switch from TextBlob to VADER or BERT.
- Make use of a more extensive real-world dataset that includes real customer feedback.
- Use Streamlit to create an interactive web application that predicts sentiment in real time.
- To manage reviews in several languages, provide multilingual support.

Project completed as part of portfolio development | Tools: Python, TextBlob, Pandas, Matplotlib